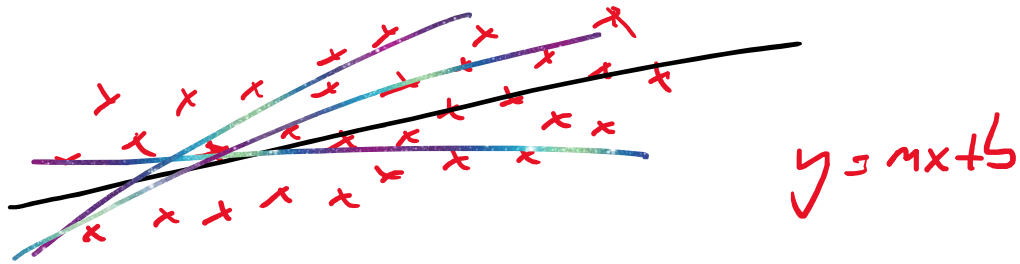


- ⇒ Supervised learning using Regression:
- ↳ Used to predict continuous values
 - ↳ Uses a straight line equation.
 - ↳ It is a parametric model.



Parameters

$$\hat{y} = \beta_0 + \beta_1 x_i$$

Linear Reg Formula

Dependent Variable

Intercept (Constant)

Slope (Regression Coeff)

Independent Variable (Explanatory)

Height (cm) (x)

Weight (kgs) (y)

160

72

171

76

182

77

180

83

154

76

 $\Rightarrow$

$$\beta_1 = \frac{\sum (x_i - \bar{x}) \cdot (y_i - \bar{y})}{\sum (x_i - \bar{x})^2}$$

$$\beta_0 = \bar{y} - (\beta_1 \cdot \bar{x})$$

$\Rightarrow$ Step 1: Find the mean of x & y

$\Rightarrow$ Step 2: Find the values of data for

β_0 & β_1

$$\Rightarrow \bar{x} = \frac{\sum x_i}{N}$$

$$\bar{y} = \frac{\sum y_i}{N}$$

$$\bar{x} = \frac{160 + 171 + 182 + 180 + 154}{5}$$

$$\bar{y} = \frac{72 + 76 + 77 + 83 + 76}{5}$$

$$\bar{x} = 169.4$$

$$\bar{y} = 76.8$$

$$\bar{x} = \underline{\underline{169.4}}$$

$$\bar{y} = \underline{\underline{76.8}}$$

x_i	y_i	$x_i - \bar{x}$	$y_i - \bar{y}$	$(x_i - \bar{x})^2$	$(y_i - \bar{y})^2$	$(x_i - \bar{x}) \cdot (y_i - \bar{y})$
160	72	-9.4	-4.8	88.36	23.04	45.12
171	76	1.6	-0.8	2.56	0.64	-1.28
182	77	12.6	0.2	158.76	0.04	2.52
180	83	10.6	6.2	112.36	38.44	65.72
159	76	-15.4	-0.8	237.16	0.64	12.32
		$\Sigma = 599.2$		$\Sigma = 52.8$	$\Sigma = 124.4$	

$$(x_1 - \bar{x}) = 160 - 169.4 = -9.4 \quad (y_1 - \bar{y}) = 72 - 76.8$$

$$(x_2 - \bar{x}) = 171 - 169.4 = 1.6 \quad (y_2 - \bar{y}) = 76 - 76.8$$

Coming back to formulas:

$$\beta_1 = \frac{\Sigma (x_i - \bar{x}) \cdot (y_i - \bar{y})}{\Sigma (x_i - \bar{x})^2} = \frac{124.4}{599.2}$$

$$= \underline{\underline{0.208}}$$

$$\begin{aligned}
 \beta_0 &= \bar{y} - (\beta_1 \cdot \bar{x}) \\
 &= 76.8 - (0.208 \times 169.4) \\
 &= \underline{\underline{41.5648}}
 \end{aligned}$$

$$\therefore \hat{y} = \beta_0 + \beta_1 \cdot x_i$$

$$\hat{y} = (41.5648) + (0.208 \cdot x_i)$$

$\Rightarrow$ Assume the given height of person is 176 cms. Predict the weight of the person.

$$\hat{y} = 41.5648 + (0.208 \times 176)$$

$$\hat{y}_{176} = 78.172 \text{ kgs}$$

$\Rightarrow$ ANOTHER APPROACH :

Using Karl Pearson's Co-efficient of Correlation.

$$r = \underline{\underline{\frac{\sum (x_i - \bar{x}) \cdot (y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \cdot \sum (y_i - \bar{y})^2}}}}$$

$$\sqrt{\sum (x_i - \bar{x})^2 \cdot \sum (y_i - \bar{y})^2}$$

$$r = \frac{124.4}{\sqrt{(599.2) \times (62.8)}} = \underline{\underline{0.6413}}$$

Next Step: Finding standard deviation of x & y

$$S_x = \sqrt{\frac{\sum (x_i - \bar{x})^2}{N-1}} \quad \& \quad S_y = \sqrt{\frac{\sum (y_i - \bar{y})^2}{N-1}}$$

$$S_x = \sqrt{\frac{599.2}{4}} = \boxed{12.24} \quad \& \quad \sqrt{\frac{62.8}{4}}$$

$$S_y = \boxed{3.96}$$

Now:

$$\beta_1 = r \cdot \frac{S_y}{S_x} = 0.6413 \times \frac{3.96}{12.24} = \underline{\underline{0.208}}$$

$$\begin{aligned} \beta_0 &= \bar{y} - (\beta_1 \cdot \bar{x}) = 76.8 - (0.208 \times 167.4) \\ &= \underline{\underline{41.5648}} \end{aligned}$$

$\Rightarrow$ Cost Function: (Loss Fn) :

Tracks the loss information from
a model.

i) MAE ii) MSE iii) RMSE

$\Rightarrow$ MSE - Mean Squared Error

$$\text{Error} = y_i - \hat{y}_i$$

$$\text{Sum of Errors} = \sum (y_i - \hat{y}_i)$$

$$\text{Sum of Squared Error} = \sum (y_i - \hat{y}_i)^2$$

$$\text{Mean Squared Error} = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2$$

$$\hat{y} = \beta_0 + \beta_1 \cdot x_i$$

$$= 41.5648 + (0.200 \times x_i)$$

$$= 71.5648 + 0.208 \times x_1$$

x_i	y_i	$\hat{y}_i$	
160	72	74.8448	$\left[\begin{array}{l} 71.5648 + (0.208 \times x_1) \\ 71.5648 + (0.208 \times 160) \end{array} \right]$
171	76	77.1328	$\left[\begin{array}{l} 71.5648 + (0.208 \times x_2) \\ 71.5648 + (0.208 \times 171) \end{array} \right]$
182	77	79.4208	
180	83	79.0048	
159	76	73.5968	

$$MSE = \frac{1}{5} \cdot \left[(72 - 74.8448)^2 + (76 - 77.1328)^2 + (77 - 79.4208)^2 + (83 - 79.0048)^2 + (76 - 73.5968)^2 \right]$$

$$MSE = \frac{8.092 + 1.283 + 5.860 + 15.562 + 5.775}{5}$$

$$MSE = \underline{\underline{7.3944}}$$

$\Rightarrow$ Performance Metric Analysis : Accuracy

$\hookrightarrow$ Accuracy for Regression models is said to be "Goodness of Fit".

↳ The metric for G-O-F is R^2 Score.

↳ Also known as Coefficient of Determination

$$\Rightarrow R^2/r^2 = \frac{\text{Explained Variation}}{\text{Total Variation}}$$

$$R^2 = 1 - \frac{SSE}{SST} \quad \text{or} \quad \frac{SSR}{SST}$$

SSE - Sum of Squared Error

SSR - Sum of Squared Regressors

SST - Sum of Squared Total

$$SSE = \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

$$SSR = \sum_{i=1}^n (\hat{y}_i - \bar{\hat{y}})^2$$

$$SST = \sum_{i=1}^n (y_i - \bar{y})^2$$

$$SST = \sum_{i=1}^n (y_i - \bar{y})^2$$

$$R^2 = 1 - \frac{SSE}{SST} = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}$$

$$R^2 = \frac{SSR}{SST} = \frac{\sum_{i=1}^n (\hat{y}_i - \bar{y})^2}{\sum_{i=1}^n (y_i - \bar{y})^2}$$

$$\bar{y} = 76.8$$

$$\begin{aligned} \bar{\hat{y}} &= 74.8448 + 77.1328 + 79.4208 + 79.0048 \\ &\quad + 73.5958 \\ &\quad \underline{\hspace{10em}} \\ &\quad \quad 5 \end{aligned}$$

$$\bar{\hat{y}} = \underline{\underline{76.8}}$$

$$SSE = \{(72 - 74.8448)^2 + (76 - 77.1328)^2 + (77 - 79.4208)^2 + (83 - 79.0048)^2 + (73 - 73.5958)^2\}$$

$$\begin{aligned} & (77.754208)^2 + (83 - 75.0048)^2 + \\ & (76 - 73.5968)^2 \end{aligned}$$

$$SSE = 36.9733$$

$$\begin{aligned} SSR &= (74.8448 - 76.8)^2 + (77.1328 - 76.8)^2 + \\ & (75.4208 - 76.8)^2 + (75.0048 - 76.8)^2 + \\ & (73.5968 - 76.8)^2 \end{aligned}$$

$$\sum (\hat{y}_i - \bar{y})^2$$

$$SSR = 25.9237$$

$$\begin{aligned} SST &= \sum (y_i - \bar{y})^2 \\ &= \underline{\underline{62.8}} \end{aligned}$$

$$\therefore R^2 = 1 - \frac{SSE}{SST} = 1 - \frac{36.9733}{62.8}$$

$$= 0.41125 \approx \underline{\underline{41.125\%}}$$

→ Another Method :

$$\text{Coeff of Determination} = \left(\text{Karl Pearson's Coeff of Correlation} \right)^2$$

$$r^2 = R \times R \approx (r \times r)$$

$$= (0.643)^2$$

$$= 41.125\% \approx \underline{\underline{0.41125}}$$